

Network Administration and Management Lab. Work

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Part I

Course Information

SYLLABUS

Course info

Class	Day	Time	Location
Lecture RI5B1	Tuesday	12:15 pm - 3:55 pm	Lab HS104
Lecture RI5B2	Wednesday	12:15 pm - 3:55 pm	Lab HS104

Learning objectives

By the end of the semester, you will be able to...

- install and configure an Operating System (Ubuntu Server).
- install and configure web-related server services such as DNS Server and Web Server.
- integrate the installation of network service technologies within a simple to medium network topology.
- install and configure email-related server services such as Mail Server and Mail Client.
- install and demonstrate Webmin features for network administration and management needs.

Lectures and labs

Lab Units

1. Instalasi Ubuntu Server
2. Samba Server
3. FTP Server Ubuntu
4. FTP Server Windows
5. Web Server
6. DNS Server
7. Proxy Server
8. SARG
9. Mail Server
10. Webmin

Assessment

Assessment for the course is comprised of six components: application exercises, homework assignments, labs, exams, projects, and teamwork.

Labs

Each lab unit requires a **Lab Report** submitted according to the rubric outlined in the Grading section below.

Grading

Grading Criteria

Grade Component	
Percentage	
Lab Reports	40%
Participation	10%
Midterm Exam	20%
Project	30%

Lab Report Grading Rubric

No	Category	Max Score	Criteria
1	Experiment Results	40	Completeness and appropriateness of screenshots for each stage.

No	Category	Max Score	Criteria
2	Discussion	40	Depth of discussion on the lab results. Explain the purpose or effect of each configuration. Explain any findings (problems and solutions) during the lab execution. If there are no specific findings, note important factors that require attention for the lab unit to be successful.

No	Category	Max Score	Criteria
3	Conclusion	10	Aligns with the lab objectives and is informative.
4	Formatting	10	Neatness of the report, including font usage, indentation, paragraph alignment, and image clarity.
-	Total Score	100	

Course policies

0.0.1 1) Academic integrity

TL;DR: Don't cheat!

All students must adhere to the Universitas Gadjah Mada (UGM) Code of Ethics. As a student of the Internet Engineering Technology (Teknologi Rekayasa Internet) program, you commit to upholding a culture of honesty, fairness, and accountability by pledging:

- I will not lie, cheat, plagiarize, or steal in my academic endeavors.
- I will conduct myself honorably.
- I will act if this standard is compromised.

Your Responsibilities Regardless of the class format, you must follow all UGM academic policies. This means:

- Submitting your own original work (including programming code).
- Properly citing all sources.
- Strictly following guidelines for group projects.

Ignoring these requirements is a serious violation. If you have any questions about these policies, please contact your Academic Advisor (DPA) or the Head of the Internet Engineering Technology Program (Sekolah Vokasi UGM).

! Important

TipSen is a form of academic dishonesty. Students who submit TP will receive a zero for the assignment and may face further disciplinary action.

0.0.2 2) Late work policy

The due dates for assignments are there to help you keep up with the course material and to ensure the teaching team can provide feedback within a timely manner. We understand that things come up periodically that could make it difficult to submit an assignment by the deadline.

- There is no late **weekly report** accepted for application exercises, since these are designed to help you prepare for labs and homework.
- The late work policy for exams will be provided with the exam instructions.
- The late work policy for the project will be provided with the project instructions.

Lab Report Submission — Late Penalty

Lateness Penalty	
1 hour	-10 points
– 1 day late	
1 day late	-20 points
– 1 week late	

0.0.3 3) Attendance policy

Maximum attendance is 75% of the total number of classes. Students who do not meet this requirement will receive a failing grade for the course.

Important dates

- 18 August - 2 October 2026: Lecture 1
- 5 October - 16 October 2026: Mid Exam
- 19 October - 4 December 2026: Lecture 2
- 7 December - 18 December 2026: Final Exam

Five tips for success

Your success on this course depends very much on you and the effort you put into it. The course has been organized so that the burden of learning is on you. Your LAs and I will help you by providing you with materials and answering questions and setting a pace, but for this to work you must do the following:

1. Complete all the preparation work before class.

2. Ask questions. As often as you can. In class, out of class. Ask me, ask the LAs, ask your friends, ask the person sitting next to you. This will help you more than anything else. If you get a question wrong on an assessment, ask us why. If you're not sure about the homework, ask. If you hear something on the news that sounds related to what we discussed, ask. If the reading is confusing, ask.
3. Do the readings.
4. Do the homework and lab. The earlier you start, the better. It's not enough to just mechanically plow through the exercises. You should ask yourself how these exercises relate to earlier material, and imagine how they might be changed (to make questions for an exam, for example.)
5. Don't procrastinate. If something is confusing to you in Week 2, Week 3 will become more confusing, Week 4 even worse, and eventually you won't know where to begin asking questions. Don't let the week end with unanswered questions. But if you find yourself falling behind and not knowing where to begin asking, come to office hours, and let me help you identify a good (re)starting point.

TEAM

Lecturers

Dr. Ir. Sahirul Alam, S.T., M.Eng. (he/his) is an Indonesian academic, researcher, and lecturer at Universitas Gadjah Mada in Yogyakarta, Indonesia. He currently serves as a faculty member in the Department of Electrical Engineering and Informatics within the Vocational College. In addition to his teaching duties, he holds the management position of Secretary for the Applied Bachelor in Internet Engineering Technology study program.

Dr. Alam's research interests lie at the intersection of advanced telecommunications, network optimization, and computational engineering. His academic work primarily focuses on wireless communication systems, RADAR signal processing, optimization methodologies, and Internet of Things (IoT).

Achmad Solaeman, S.Tr.T., M.Eng. (he/his) is an academic and lecturer in the Internet Engineering Technology program at the Vocational College, UGM. His expertise lies in Distributed Artificial Intelligence and Network Security, with active research in deep learning applications. He is dedicated to driving technological innovation and shaping future professionals in the networking and AI domains.

Lab assistants

Class	NIU	Name	Email
RI5B1	XXXXXX	Rich	as@mail.ugm.ac.id
RI5B1	XXXXXX	Rich	asd@mail.ugm.ac.id
RI5B2	535867	Asisten1	as@mail.ugm.ac.id
RI5B2	539428	As	asisten2@mail.ugm.ac.id
RI5A1	5	Assistant Name 5	assistant5@mail.ugm.ac.id
RI5A1	6	Assistant Name 6	assistant6@mail.ugm.ac.id
RI5A2	556548	Mina	asdw@mail.ugm.ac.id
RI5A2	555596	Aditya	asdw@mail.ugm.ac.id

Part II

Weekly materials

CHAPTER I UBUNTU SERVER INSTALLATION

1.1 Practical Objectives

- a. Students are able to install Ubuntu Server.
- b. Students understand and are able to practice basic commands in the Ubuntu Server environment.

1.2 Theoretical Background of Ubuntu Server

Ubuntu Server is a Linux-based server operating system based on the Ubuntu distribution, developed by Canonical Ltd. Ubuntu Server is specifically designed for use in server environments and network infrastructure. Ubuntu Server is the version of the Ubuntu operating system without a graphical user interface (headless), with a primary focus on performance and efficiency for server-related tasks [1].

The following are some important points about Ubuntu Server:

1. Open Source: Ubuntu Server is open-source, meaning that its source code can be freely accessed, used, modified, and distributed by anyone. As a result, many communities and organizations can contribute to the development and improvement of the operating system.
2. Stability and Security: Ubuntu Server is supported by a large and active community and is reviewed by security experts. Stable versions are released regularly with Long-Term Support (LTS), providing security updates and bug fixes for an extended period, typically up to five years after release.
3. APT Package Management: Like other Linux operating systems, Ubuntu Server uses the Advanced Package Tool (APT) as its package management system. This allows administrators to easily install, remove, and manage software and additional packages required by the server system.
4. Command-Line Interface (CLI): Ubuntu Server operates through a CLI, allowing administrators to manage the system remotely or through Secure Shell (SSH). The absence of a graphical interface helps reduce resource usage and improve security.
5. Cloud Support: Ubuntu Server is well known for its support for cloud computing environments. This server version is widely used by cloud service providers such as Amazon Web Services (AWS), Google Cloud Platform (GCP), and Microsoft Azure.
6. Virtualization Support: Ubuntu Server supports virtualization technologies such as Kernel-based Virtual Machine (KVM) and Docker containers. This allows users to create and manage virtual machines and containers easily and efficiently.
7. Access to Ubuntu Repositories: Ubuntu Server provides access to extensive Ubuntu repositories for downloading and installing various server software, including web applications, databases, email servers, and many others.
8. Documentation and Community Support: Ubuntu has a large and active community that can provide assistance through forums, discussion groups, and other online resources. In addition, paid support options are available from Canonical Ltd. for business and enterprise users.

Ubuntu Server has become a popular choice for building and managing server infrastructure due to its stability, security, and support for modern technologies such as cloud computing and virtualization. It provides powerful tools for operating servers reliably and efficiently [2]. The hands-on steps for installing and configuring it are covered in Section 1.3.

1.3 Ubuntu Server Practicum Steps

This guide walks through installing Ubuntu Server in VirtualBox and configuring its network adapter so the machine can be reached from outside VirtualBox.

1.3.1 Installing Ubuntu Server in VirtualBox

i What you'll need

An internet connection is required during installation, since packages are fetched while Ubuntu Server sets itself up.

1. **Download the Ubuntu Server installer image** from ubuntu.com/download/server. Choose the version that matches your computer's specifications. **Ubuntu Server 24** is the recommended version for this practicum.
2. **Install Ubuntu Server through VirtualBox.**

! Recommended virtual machine specifications

Setting	Recommended value
Base Memory	8 GB
Processors	2
Storage (VDI)	30 GB

Use a **different drive** from the one hosting your host operating system to store the VM folder!

1.3.2 Configuring the Network Adapter for External Access

The default network adapter in VirtualBox is isolated from the outside world. Adding a second, host-only adapter lets you reach the Ubuntu Server VM from your host machine, configured as shown in Figure 1.1.

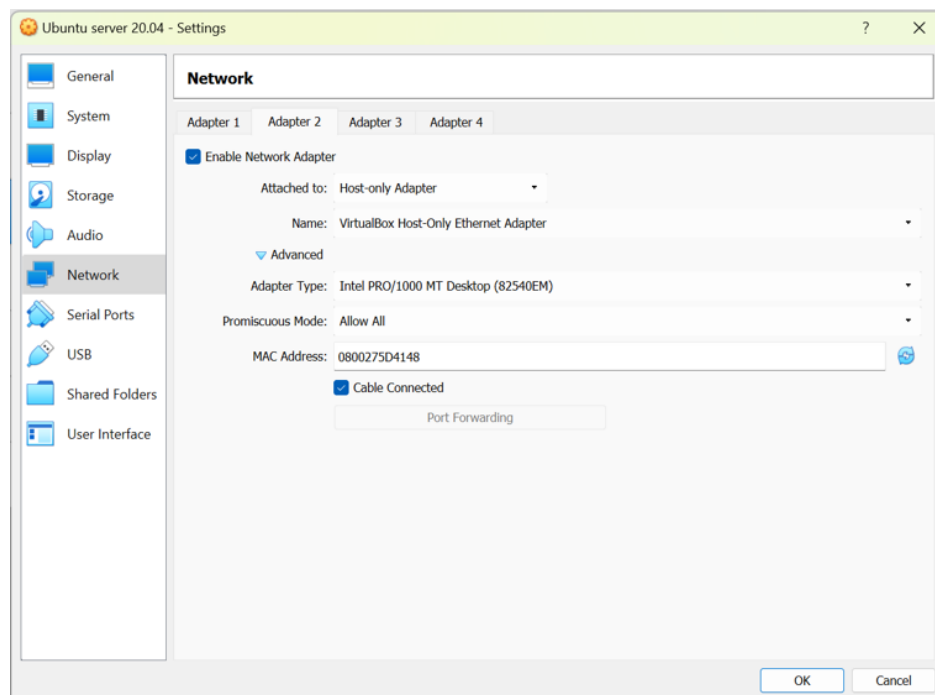


Figure 1.1: Setting on VMWare via Adapter 2

! Before you start

Power off the Ubuntu Server VM you just installed before changing its network settings.

1. **Enable a host-only adapter.**

In the VM's **Settings**, open the **Network** menu and select the **Adapter 2** tab. Then:

- Set **Attached to:** → Host-only Adapter
- Set **Promiscuous Mode:** → Allow All

2. **Power the VM back on** and edit the Netplan configuration [3].

Create a configuration file with:

```
sudo nano /etc/netplan/01-netconf.yaml
```

Add the following content to the file:

```
network:
  version: 2
  renderer: networkd
  ethernets:
    enp0s8:
      dhcp4: yes
```

Save and exit the editor by pressing Ctrl+X, then Y, then Enter.

3. **Apply the new network configuration.**

In the Ubuntu Server terminal, run:

```
sudo netplan apply
```

4. **Verify the IP address.**

Check that Ubuntu Server has received an IP address on the **enp0s8** adapter. You can confirm this with:

```
ip a
```

i Example result

In the example used for this guide, the Ubuntu Server IP address to be used in the next parts of the practicum is:

192.168.56.104

Figure 1.2 shows an example of this output in the Ubuntu Server terminal.

! Keep this address handy

The IP address obtained on the **enp0s8** adapter (e.g. 192.168.56.104) will be used to access this Ubuntu Server instance in later steps of the practicum. Write it down before continuing.

1.4 Exercise

Try the commands listed in Table 1.2, explain their functions, and provide screenshots of your results. Complete a total of 15 commands.

```

alam@alam:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:3a:ea:f8 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86366sec preferred_lft 86366sec
    inet6 fe80::a00:27ff:fe3a:eaf8/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:55:f5:18 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.104/24 brd 192.168.56.255 scope global dynamic enp0s8
        valid_lft 565sec preferred_lft 565sec
    inet6 fe80::a00:27ff:fe55:f518/64 scope link
        valid_lft forever preferred_lft forever

```

Figure 1.2: Ubuntu Server

Table 1.2: Essential Linux Commands

#	Command	#	Command	#	Command	#	Command
1	ls	22	ifconfig	43	lsof	64	parted
2	pwd	23	ip	44	dig	65	wc
3	cd	24	wget	45	nslookup	66	ls
4	clear	25	curl	46	du	67	nmap
5	mkdir	26	apt	47	tree	68	dmesg
6	mv	27	apt-get	48	ss	69	chattr
7	cp	28	yum	49	partx	70	usermod
8	rmdir	29	dnf	50	uptime	71	free
9	touch	30	rpm	51	tr	72	cron
10	cat	31	alias	52	ping	73	mysql
11	echo	32	dd	53	zcat	74	sdiff
12	less	33	top	54	xargs	75	history
13	tar	34	useradd	55	rm	76	netstat
14	grep	35	sleep	56	stat	77	sftp
15	head	36	screen	57	who	78	tcpdump
16	tail	37	pv	58	locate	79	scp
17	sort	38	fgrep	59	host	80	rsync
18	ps	39	dir	60	find	81	fsck
19	kill	40	egrep	61	fuser	82	bc
20	df	41	ssh	62	at	83	chage
21	chown	42	fd	63	fdisk	84	ffmpeg

CHAPTER II SAMBA SERVER

2.1 Practical Objectives

- Students are able to install Samba Server.
- Students are able to modify Samba configuration to manage user access rights and shared resources.
- Students are able to practice file sharing between Ubuntu and Windows using Samba.

2.2 Theoretical Background of Samba Server

Samba is a free and open-source software that provides an implementation of the SMB/CIFS protocol for non-Windows operating systems such as Linux, Unix, and macOS [4]. A Samba server allows these operating systems to function as a file and print server in a Windows network environment, enabling Windows users to share and access files and printers from computers running non-Windows operating systems, as illustrated in Figure 2.1. Samba provides features such as authentication, authorization, encryption, and access rights management to ensure the security and privacy of shared data. It can also be integrated with directory services such as Active Directory to simplify user management in large network environments [5].

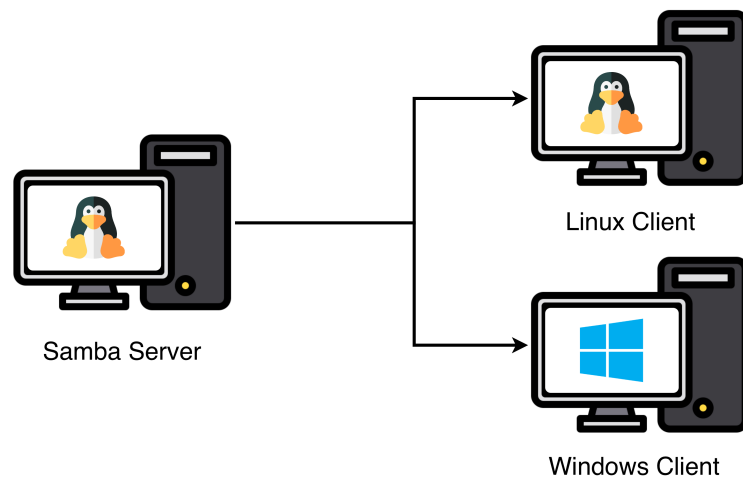


Figure 2.1: Illustration of Samba’s ability to share resources across operating systems

Samba operates using Server Message Block (SMB) protocol — a network protocol used to share files, printers, and other network resources between Windows or Unix computers [6]. SMB was originally developed by Microsoft as part of LAN Manager in the 1980s and has since become the de facto standard for file and printer sharing in Windows networks. SMB is a client-server protocol: the client sends requests to the server, and the server responds by delivering data or performing the requested operation, as illustrated in Figure 3.1. Data exchange is secured using encryption, authentication, and authorization.

Samba consists of two programs running in the background:

- **SMBD** — the file server; spawns a new process for each active client.
- **NMBD** — converts NetBIOS computer names to IP addresses and monitors shared resources on the network.

SMBD behavior is controlled through `/etc/samba/smb.conf` [7]. Samba can be configured as a

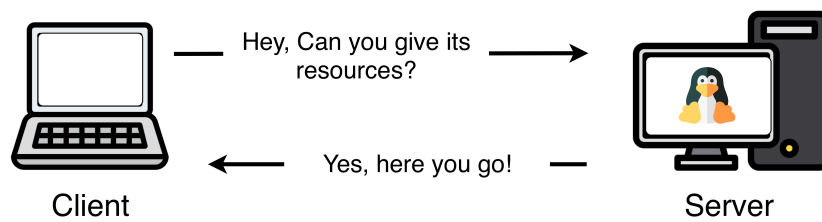


Figure 2.2: Illustration of the client-server protocol

file server, print server, or domain controller.

Uses of Samba Server

- Share one or more directory trees across the network.
- Centrally manage printers, print settings, and drivers for Windows clients.
- Assist clients with network browsing.
- Authenticate clients logging in to a Windows domain.

Advantages of Samba

- Free and open-source — no licensing cost.
- Available on multiple platforms for file sharing services.
- Easy to configure and administer.
- Connects directly to the network with minimal issues.
- Highly reliable — errors are uncommon unless the server has hardware problems.

Key Features of Samba

1. **User Authentication** — Users must log in before accessing Samba services. Samba accounts are managed with `smbpasswd`.
2. **Write Access Control** — Write permissions can be set per-share using `read only` or `writable` directives.
3. **Network Access Control** — Access can be restricted by allowing or blocking specific IP addresses.

2.3 Samba Server Installation

! Before you begin

These steps assume Ubuntu Server is already installed and running. If not, complete the Ubuntu Server installation on Chapter 1.

2.3.1 1. Installation

Step 1 — Log in as root

Log in to your Ubuntu Server terminal as root (or prefix each command with `sudo`).

Step 2 — Update the package list

```
apt update
```

Step 3 — Install Samba

```
apt install samba
```

①

- ① An active internet connection is required — packages are downloaded during installation.

Continue the installation until complete.

Step 4 — Verify the installation

```
systemctl status smbd
```

①

- ① **smbd** is the main Samba file-server daemon. A status of **active (running)** confirms a successful install.

If successful, the output will show **active (running)**, as shown in Figure 2.3.

```
● smbd.service - Samba SMB Daemon
   Loaded: loaded (/lib/systemd/system/smbd.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2022-08-30 01:37:05 UTC; 27s ago
     Docs: man:smbd(8)
           man:samba(7)
           man:smb.conf(5)
  Process: 3485 ExecStartPre=/usr/share/samba/update-apparmor-samba-profile (code=exited, status=
 Main PID: 3504 (smbd)
   Status: "smbd: ready to serve connections..."
    Tasks: 4 (limit: 1066)
   Memory: 9.1M
    CGroup: /system.slice/smbd.service
            └─3504 /usr/sbin/smbd --foreground --no-process-group
              └─3506 /usr/sbin/smbd --foreground --no-process-group
                └─3507 /usr/sbin/smbd --foreground --no-process-group
                  └─3508 /usr/sbin/smbd --foreground --no-process-group
```

Figure 2.3: Samba running successfully

If the status shows **inactive (dead)** (Figure 2.4), try starting Samba manually:

```
service smbd start
```

If it still does not start, reinstall Samba.

```
● smbd.service - Samba SMB Daemon
   Loaded: loaded (/lib/systemd/system/smbd.service; enabled; vendor preset: enabled)
   Active: inactive (dead)
     Docs: man:smbd(8)
           man:samba(7)
           man:smb.conf(5)
```

Figure 2.4: Samba inactive — installation may have failed

2.3.2 2. Firewall Setup

Step 5 — Check firewall status

```
ufw status
```

If the firewall is not active, enable it first:

```
ufw enable
```

Allow Samba traffic through the firewall

```
ufw allow Samba
```

①

- ① This opens UDP ports 137–138 and TCP ports 139 and 445, which Samba needs to communicate with Windows clients.

 Do not skip this step

Without allowing Samba through the firewall, Windows clients will not be able to connect to the shared folders.

2.4 Exercise

2.4.1 Setting Up File Sharing

Step 1 — Create the shared directories

```
mkdir /home/share1  
mkdir /home/share2
```

①

- ① Both directories will be the root of each Samba share.

Create an empty test file inside each directory:

```
touch /home/share1/testfile1.txt  
touch /home/share2/testfile2.txt
```

Step 2 — Set directory permissions

```
chmod 777 /home/share1  
chmod 777 /home/share2
```

①

- ① 777 grants read, write, and execute permission to all users. For production use, more restrictive permissions are recommended.

Step 3 — Create Samba users

```
useradd user1  
useradd user2
```

①

- ① This creates a Linux system user. A separate Samba password is still required in the next step.

Step 4 — Set Samba passwords

```
smbpasswd -a user1
```

①

- ① The `-a` flag adds a new Samba user. Enter the password twice when prompted.

Repeat for `user2`:

```
smbpasswd -a user2
```

Step 5 — Configure access rights in `smb.conf`

Open the Samba configuration file:

```
nano /etc/samba/smb.conf
```

Scroll to the bottom and add the following blocks:

```
[share1]  
path = /home/share1 # <1>  
valid users = user1, user2 # <2>
```

```
read list = user2 # <3>
write list = user1 # <4>
browseable = yes

[share2]
path = /home/share2
valid users = user1, user2
read list = user1
write list = user2
browseable = yes
```

- ① The directory on the server that will be shared.
- ② Only these users may connect to this share.
- ③ **user2** can only read files in **share1**.
- ④ **user1** can read and write files in **share1**.

Save and exit by pressing Ctrl+X, then Y, then Enter.

Folder	user1	user2
share1	Read + Write	Read only
share2	Read only	Read + Write

Step 6 — Restart Samba

```
systemctl restart smbd
```

①

- ① Configuration changes only take effect after a restart.

2.4.2 Testing File Sharing from Windows

Step 1 — Find the Ubuntu Server IP address

```
ip a
```

①

- ① Look for the IP address on the **enp0s8** adapter (the host-only adapter configured earlier).

Note the IP address (e.g., 192.168.56.101).

Step 2 — Open the shares in File Explorer

On your Windows machine, open **File Explorer** and type the following in the address bar:

```
\\192.168.56.101
```

Confirm that both **share1** and **share2** are visible, as shown in Figure 2.5.

Step 3 — Test access as user1

Click **share1** and log in as **user1** when prompted. Check each of the following:

- Can you view the contents of **share1**?
- Can you create a new file inside **share1**?
- Can you copy a file from **share1** to local Windows storage?

Then navigate to **share2** (still logged in as **user1**) and repeat the same checks.

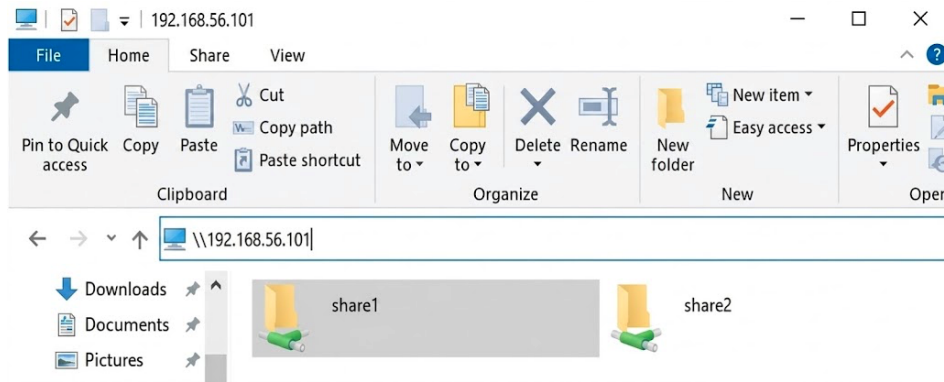


Figure 2.5: Samba shares visible in Windows File Explorer

💡 Expected result for user1

- **share1** → can view, create, and copy files
- **share2** → can view and copy files, but **cannot** create or modify files

Step 4 — Disconnect and switch to user2

Open **Command Prompt** on Windows and check the active connection:

```
net use
```

Disconnect it (replace \\name with the connection name shown above):

```
net use \\name /delete # <1>
```

- ① Disconnecting is necessary — Windows caches credentials, so the old session must be cleared before logging in as a different user.

Step 5 — Test access as user2

Connect again to \\192.168.56.101, log in as **user2**, and check both **share1** and **share2**.

💡 Expected result for user2

- **share1** → can view and copy files, but **cannot** create or modify files
- **share2** → can view, create, and copy files

CHAPTER III FTP SERVER

3.1 Practical Objectives

- Students are able to install an FTP Server on Ubuntu.
- Students are able to modify the FTP configuration to manage user access rights and shared resources.
- Students are able to practice file sharing and file transfer using FTP.

3.2 Theoretical Background of FTP Server

FTP (File Transfer Protocol) is a standard communication protocol used to transfer files between a server and a client [8]. FTP consists of two main components:

- **FTP Server** — a program running on the server that provides access to the files stored on it.
- **FTP Client** — a program running on the client computer, used to access files on the FTP server.

FTP uses a connection-based model, meaning a connection must be established before any data transfer can begin, as illustrated in Figure 3.1.

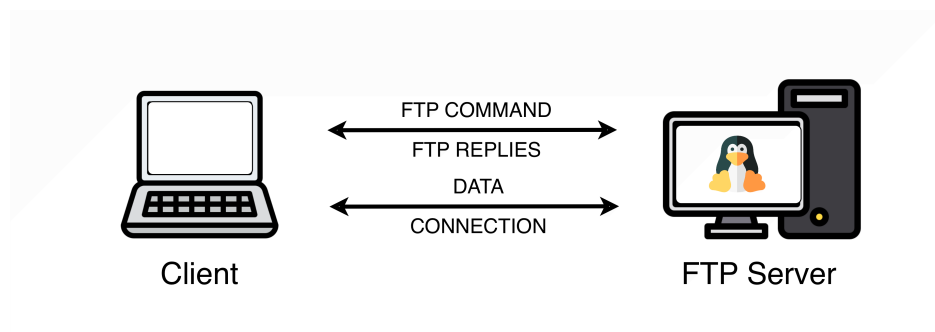


Figure 3.1: FTP connection modes

There are two types of connection modes used in FTP:

- **Active mode** — the FTP server opens a data connection to the client.
- **Passive mode** — the client opens a data connection to the server. This mode is generally preferred when the client is behind a firewall or NAT.

FTP Variants

- **FTPS** — standard FTP extended with encryption, supporting Transport Layer Security (TLS) and Secure Sockets Layer (SSL).
- **SFTP** — uses Secure Shell (SSH) as the underlying transport; not related to standard FTP despite the name.
- **Trivial FTP (TFTP)** — the simplest variant, designed for lightweight file transfers to/from a remote server without authentication.
- **Simple FTP** — complexity between TFTP and standard FTP; rarely used and largely obsolete.

3.3 FTP Server Installation

! Before you begin

These steps assume Ubuntu Server is already installed and running. If not, complete the Ubuntu Server installation practicum first.

3.4 Installation

Step 1 — Log in as root

Log in to your Ubuntu Server terminal as root (or prefix each command with `sudo`).

Step 2 — Update the package list

```
apt update
```

Step 3 — Install vsftpd

```
apt install vsftpd
```

①

① **vsftpd** (Very Secure FTP Daemon) is the FTP server package used in this practicum [9]. Continue until the installation is complete.

Step 4 — Verify the installation

```
systemctl status vsftpd
```

①

① A status of **active (running)** confirms a successful installation.

If successful, the output will show **active (running)**.

3.5 Firewall Setup

Step 5 — Check the firewall status

```
ufw status
```

If the firewall is not active, enable it first:

```
ufw enable
```

Allow FTP traffic through the firewall

```
ufw allow ftp
```

①

① This opens TCP port 21, which FTP uses for its control connection.

⚠ Do not skip this step

Without allowing FTP through the firewall, Windows clients will not be able to connect to the FTP server.

3.6 Testing

Step 6 — Test FTP access from Windows

On your Windows machine, open **File Explorer** and type the server's IP address in the address bar, prefixed with `ftp://`:

`ftp://192.168.56.101`

An FTP authentication window should appear, as shown in Figure 3.2.

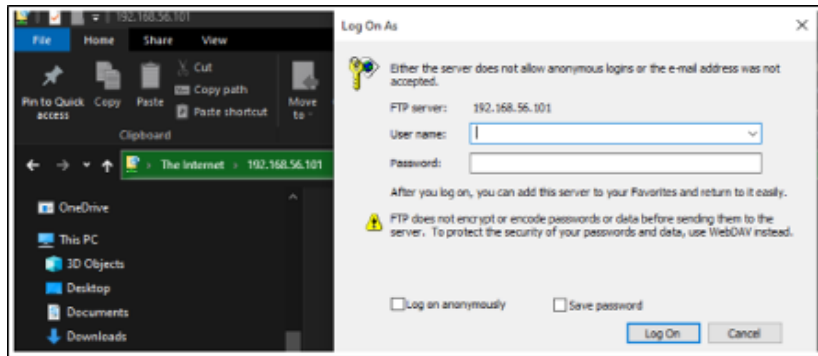


Figure 3.2: FTP authentication window in Windows File Explorer

3.7 Exercise

3.7.1 1. Creating an FTP User

a. Add a new FTP user

An FTP user owns the home directory that will be used as the FTP shared folder. Create the user before setting up the folder:

```
adduser alamftp
```

①

① Replace `alamftp` with any username you prefer.

Enter a password when prompted. For the subsequent fields (Room Number, etc.), press Enter to skip them.

b. Restart the FTP service

```
systemctl restart vsftpd
```

c. Test login from Windows

Open **File Explorer** on Windows and navigate to `ftp://192.168.56.101`. Log in with the FTP user credentials you just created and confirm that the login succeeds.

! Login fails?

If login is denied, the firewall may be blocking the connection. Temporarily disable it and try again:

```
ufw disable
```

3.7.2 2. Creating an FTP Folder

a. Verify the user's home directory

Confirm that a home folder for the FTP user was automatically created under `/home`:

```
cd /home
ls
```

①

① The FTP user's folder (e.g. `alamftp`) should appear in the listing on Figure 3.3.

```

root@alam:/home/alam# cd /home
root@alam:/home# ls
alam  alamftp
root@alam:/home#

```

Figure 3.3: FTP user's home directory

b. Create a folder and a test file

```

mkdir /home/alamftp/folder_baru ①
touch /home/alamftp/file_baru ②

```

① Creates a new sub-folder inside the FTP user's home directory.

② Creates an empty test file for file transfer simulation.

c. Verify in Windows File Explorer

Refresh **File Explorer** on Windows and confirm that `folder_baru` and `file_baru` are now visible in the FTP share.

3.7.3 3. Blocking Users

FTP users can be blocked from accessing the server by adding their username to the FTP deny list.

a. Open the FTP user block list

```

nano /etc/ftpusers ①

```

① Any username listed in this file is denied FTP access.

b. Add the username to block

Scroll to the bottom of the file and add the username on a new line (e.g. `alamftp`). Save and exit by pressing `Ctrl+X`, then `Y`, then `Enter`.

c. Test the blocked account

Close and reopen **File Explorer** on Windows, then attempt to log in with the blocked user. Access should now be denied.

d. Remove the block before continuing

Open `/etc/ftpusers` again, delete the username you added, and save the file. The user must be unblocked before proceeding to the next exercise.

3.7.4 4. Granting Access

a. Confirm that write access is not yet enabled

In Windows **File Explorer**, try to copy a file into the FTP folder or delete an existing file. Confirm that the operation fails — write access is disabled in the default vsftpd configuration.

b. Back up the configuration file

```

cp /etc/vsftpd.conf /etc/vsftpd.conf.backup ①

```

① Always back up configuration files before editing them.

c. Enable write access

Open the configuration file:

```
nano /etc/vsftpd.conf --linenumbers
```

Find **line 31** and uncomment it by removing the leading #:

```
write_enable=YES
```

Save and exit.

d. Restart the FTP server

```
systemctl restart vsftpd.service
```

①

① Configuration changes only take effect after a restart.

e. Verify write access from Windows

In **File Explorer**, copy a file from your local computer into the FTP folder, and try deleting a file from the FTP folder. Both operations should now succeed.

CHAPTER IV FTP SERVER ON WINDOWS

4.1 Practical Objectives

- a. Students are able to install an FTP Server on Windows.
- b. Students are able to modify the FTP configuration to manage user access rights and shared resources.
- c. Students are able to practice file sharing and file transfer using FTP.

4.2 Theoretical Background of FTP Server

The first specification for FTP was published as RFC 114 on 16 April 1971, written by Abhay Bhushan, then a student at the Massachusetts Institute of Technology. The original idea behind FTP was to enable file transfers over ARPANET, the precursor to the modern internet.

As the modern internet took shape, the FTP specification went through several revisions to align with evolving network standards, including TCP/IP. In 1980, a new version of FTP was defined in RFC 765 by Jon Postel, a research scientist at the Information Sciences Institute at the University of Southern California. Five years later, FTP was redefined in RFC 959, which introduced new management capabilities — including the ability to create and delete file directories [8]. In 1997, RFC 959 was updated with additional security capabilities specified in RFC 2228, and two years later, RFC 2428 added support for the IPv6 protocol.

Common Use Cases for FTP

- **Backup** — FTP can be used by backup services or individual users to copy data from one location to a secure backup server running an FTP service.
- **Replication** — similar to backup, replication duplicates data across systems to provide higher availability and resilience; FTP can facilitate this process.
- **Data access and upload** — FTP is widely used to access shared web hosting and cloud services as a mechanism for loading data to remote systems.

FTP is a client-server protocol that relies on two communication channels between the client and server: a **command channel** for controlling the conversation, and a **data channel** for transmitting file content. A typical FTP transfer works as follows:

1. A user logs in to the FTP server (some servers allow anonymous access without credentials).
2. The client initiates a conversation with the server when the user requests a file download.
3. Using FTP, the client can upload, download, delete, rename, move, and copy files on the server.

4.3 FTP Server Installation

Windows includes a built-in FTP service through Internet Information Services (IIS). No additional software is needed — the service only needs to be enabled and configured [10]. The steps below apply to **Windows 10 and Windows 11**.

4.4 1. Enabling IIS Features

Step 1 — Create the FTP root folder

Create a new folder on your **D:** drive. This folder will serve as the root directory for the FTP site.

Step 2 — Open Windows Features

Navigate to **Control Panel → Programs and Features**, then click **Turn Windows features on or off**.

Step 3 — Enable FTP and IIS components

Expand **Internet Information Services** and check the following three items:

- **FTP Extensibility**
- **FTP Service**
- **IIS Management Console**

Click **OK** and wait for the installation to complete.

4.5 2. Configuring IIS

Step 4 — Open IIS Manager

Opening IIS Manager

- **Windows 10:** Control Panel → Administrative Tools → Internet Information Services (IIS) Manager
- **Windows 11:** Control Panel → Windows Tools → Internet Information Services (IIS) Manager

Step 5 — Add a new FTP site

In the IIS Manager window, right-click on the server name (e.g. DESKTOP-XXXXXX) and select **Add FTP Site...**

Step 6 — Site Information

In the **Add FTP Site — Site Information** window:

- **FTP site name:** enter a name, e.g. FTP_coba
- **Physical path:** select the folder you created in Step 1

Step 7 — Binding and SSL Settings

In the **Add FTP Site — Binding and SSL Settings** window:

- **IP Address:** enter the IP address of this computer
- **SSL:** select **No SSL**

Step 8 — Authentication and Authorization

In the **Add FTP Site — Authentication and Authorization Information** window:

- Under **Authentication**, check **Basic**
- Click **Finish**

4.6 3. Testing

Step 9 — Test FTP access from another computer

Find the IP address of the server. Then, from another computer acting as the client, open **File Explorer** and type the FTP address in the address bar:

ftp://192.168.x.x

4.7 Exercise

4.7.1 1. Adding an FTP User

a. Create a new Windows user account

Steps differ by Windows version

Windows 10: Start → Settings → Family & other users → Add someone else to this PC

Windows 11: Search for Computer Management → Local Users and Groups → Users → right-click → New User...

b. Set a username and password

Enter a username and password for the FTP client account.

! Windows 11 only

Uncheck **User must change password at next login** before clicking **Create**.

c. Open FTP Authorization Rules

In **IIS Manager**, double-click **FTP Authorization Rules**, then click **Add Allow Rule...**

d. Authorize the new user

In the **Add Allow Authorization Rule** window:

- Select **Specified users** and enter the username created in step b.
- Under **Permissions**, check **Read**.
- Click **OK**.

e. Test login from the client computer

From the client computer, open **File Explorer** and navigate to `ftp://192.168.x.x`. Log in with the credentials created in step b and confirm the login succeeds.

4.7.2 2. Setting FTP User Permissions

a. Create test files on the server

From the **server** computer, create a new folder and a new file inside the FTP root folder created in Step C.1.

b. Refresh the client view

Refresh **File Explorer** on the client computer so the newly created folder and file appear.

c–e. Test current permissions from the client

Attempt each of the following from the **client** computer and note whether each succeeds or fails:

Action	Expected result
Copy a file from the FTP server to local storage	Should succeed (Read is allowed)
Copy a file from local storage to the FTP server	Should fail (Write not yet allowed)
Delete a file on the FTP server	Should fail (Write not yet allowed)

f. Grant write permission

From the **server**, open **FTP Authorization Rules** in IIS Manager. Right-click the FTP client username and select **Edit**. Check both **Read** and **Write** under Permissions, then click **OK**.

g. Retest write operations

Repeat the copy-to-server and delete tests from step e. Both operations should now succeed. Explain why the behaviour changed after updating the authorization rule.

4.7.3 3. Removing the FTP Server

Complete the following steps to fully remove the FTP server and restore the system to its original state.

a. Remove the FTP site

In **IIS Manager**, right-click the FTP site created in Step C.6 and select **Remove** → **Yes**.

b. Delete the FTP user account

Steps differ by Windows version

Windows 10: Start → Settings → Family & other users → click the account name → Remove → Delete account and data

Windows 11: Computer Management → Local Users and Groups → Users → right-click the user → Delete

c. Disable IIS

Navigate to **Control Panel** → **Programs and Features** → **Turn Windows features on or off**. Uncheck **Internet Information Services** entirely and click **OK**.

d. Restart the server

Restart the computer to complete the removal.

CHAPTER V WEB SERVER

5.1 Practical Objectives

- Students are able to install a Web Server.
- Students are able to configure the Web Server display using Virtual Hosting.
- Students are able to create and manage web page sub-folders.

5.2 Theoretical Background of Web Server

A web server is a computer program that provides access to websites or web applications over the HTTP or HTTPS protocol [11]. The web server stores and delivers files such as HTML, CSS, and images to clients that request access through a web browser or web application.

Beyond the web server itself, a web system consists of four main components: the **web server**, **web client**, **web application**, and **database**. The web client is the internet user accessing the web server's services, and the web application provides the software layer that supports those services, as illustrated in Figure 5.1.

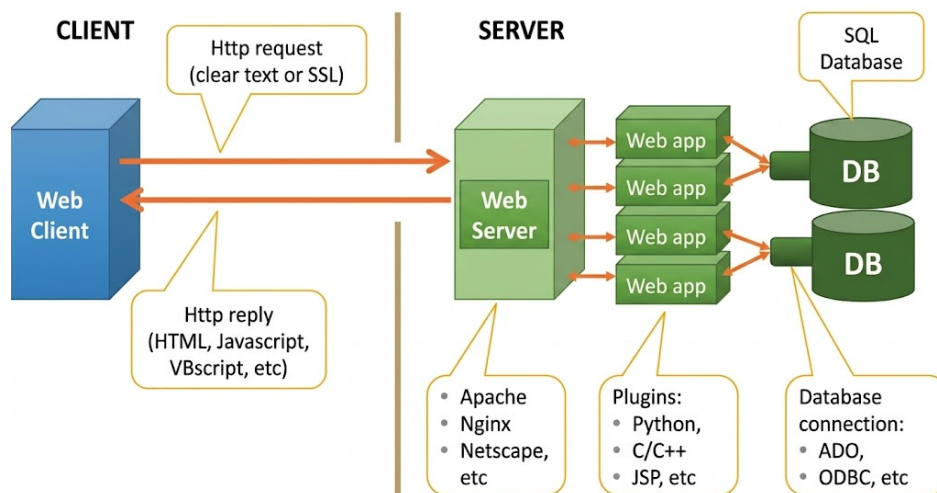


Figure 5.1: Components of a web system

HTTP (HyperText Transfer Protocol) is the network protocol used to transfer data between a web server and a web client such as a browser [12]. HTTP follows a client-server model: the web client sends an HTTP request to the server, and the server returns an HTTP response. An HTTP response contains:

- an **HTTP status code**, such as 200 OK or 404 Not Found,
- headers** carrying information about the server and the content,
- the requested **content**, such as text or images.

HTTPS

HTTPS is the secure version of HTTP, enhanced with **SSL/TLS encryption** to protect data in transit between the client and the server.

5.3 Web Server Installation

! Before you begin

These steps assume Ubuntu Server is already installed and running with network connectivity to a Windows host machine. If not, complete the Ubuntu Server installation practicum first.

5.4 1. Pre-checks

Run the following checks before installing Apache to confirm that no web server is currently running.

Step 1 — Update the package list

Log in to Ubuntu Server as root, then run:

```
apt update
```

Step 2 — Find the Ubuntu Server IP address

```
ip a
```

①

① Note the IP address on the `enp0s8` adapter (e.g. `192.168.56.103`).

Step 3 — Ping from Windows

On the Windows host, open **Command Prompt** and ping the Ubuntu Server IP address. A successful reply confirms network connectivity between the two machines.

Step 4 — Confirm no web server is running (from Windows)

If the ping succeeds, open a web browser on Windows and navigate to:

`http://192.168.56.103`

The expected result is an **Unable to connect** or **Connection timed out** message, confirming that no web server is currently active.

Step 5 — Confirm no web server is running (from Ubuntu)

If the Windows ping fails, verify from within Ubuntu Server instead:

```
apt install w3m
```

```
w3m -v http://192.168.56.103
```

①

① `w3m` is a text-based web browser useful for testing web access directly from the terminal.

The expected output is: **No such file or directory**.

5.5 2. Apache Installation

Step 1 — Install Apache

```
apt install apache2
```

①

① Apache2 is one of the most widely used open-source web servers. An active internet connection is required during installation.

Step 2 — Check Apache in the firewall app list

```
ufw app list
```

①

- ① Confirm that `Apache`, `Apache Full`, and `Apache Secure` appear in the list.

Step 3 — Enable the firewall if inactive

```
ufw status
```

①

```
ufw enable
```

②

- ① Check the current firewall status.
② Run only if the status shows `inactive`.

Step 4 — Allow Apache through the firewall

```
ufw allow 'Apache'
```

①

```
ufw status
```

- ① Grants Apache permission to receive incoming HTTP traffic on port 80.

5.6 3. Verification

Step 1 — Check that Apache is running

```
systemctl status apache2
```

①

- ① The output should show `active (running)` to confirm a successful installation.

Step 2 — Access the Apache landing page

From Windows, navigate to `http://192.168.56.103` in a browser — the default Apache landing page should now appear. Alternatively, from the Ubuntu terminal:

```
w3m -v http://192.168.56.103
```

A page containing “**Apache2 Default Page**” content confirms the server is working correctly.

5.7 Exercise

5.8 1. Virtual Hosting

Virtual Hosting allows you to serve a custom website from your own directory instead of Apache’s default root.

Step 1 — Explore the default web root

```
cd /var/www
```

①

```
ls
```

- ① Apache’s default content lives in `/var/www/html`. You will create a sibling directory for your own site.

Step 2 — Create a directory and a sample page

```
mkdir /var/www/your_site_name
```

```
nano /var/www/your_site_name/index.html
```

Step 3 — Edit `index.html`

Type the following content into the editor:

```
<html>
  <head>
    <title>Welcome to Your Site Name!</title>
  </head>
  <body>
    <h1>Success! Your Virtual Host is working.</h1>
  </body>
</html>
```

Save and exit with Ctrl+X, then Y, then Enter.

Step 4 — Update the Apache configuration

Open the default virtual host configuration file:

```
nano /etc/apache2/sites-available/000-default.conf
```

Find the DocumentRoot line, comment it out, and add a new line pointing to your directory:

```
# DocumentRoot /var/www/html
DocumentRoot /var/www/your_site_name
```

Save and exit.

Step 5 — Restart Apache and verify

```
systemctl restart apache2
```

Refresh the browser at <http://192.168.56.103> (or re-run `w3m -v http://192.168.56.103` from Ubuntu). Your custom page should now appear instead of the default Apache page.

5.9 2. Sub-folders

Sub-folders let you serve additional pages under paths such as `/about` or `/contact`.

Step 1 — Create a sub-directory and its index page

```
mkdir /var/www/your_site_name/about
nano /var/www/your_site_name/about/index.html
```

Step 2 — Edit the sub-folder index.html

```
<html>
  <head>
    <title>About - Your Site Name</title>
  </head>
  <body>
    <h1>This is the About page of Your Site Name.</h1>
  </body>
</html>
```

Save and exit.

Step 3 — Verify the new page

From the Windows browser:

<http://192.168.56.103/about>

Or from the Ubuntu terminal:

```
w3m -v http://192.168.56.103/about
```

The About page content should be displayed. Note that Apache automatically serves `index.html` when a directory path is requested — no additional configuration is needed for basic sub-folders.

CHAPTER VI DNS SERVER

6.1 Practical Objectives

- Students are able to install a DNS Server.
- Students are able to modify the DNS Server configuration so that a server's IP address can be accessed using a custom domain name.

6.2 Theoretical Background of DNS Server

DNS (Domain Name System) is a distributed database system that maps host names, machine names, and domain names to IP addresses — and conversely maps IP addresses back to host names, a process known as **reverse mapping** [13].

DNS acts as a bridge for two fundamental reasons:

- Humans find it easier to remember names than 32-bit IP addresses.
- Computers use IP addresses to communicate and interact over networks.

How DNS works:

- When a user types a URL or domain name into their browser, the browser queries the local DNS server connected to the network. If the local server does not have the answer, the request is forwarded up the DNS hierarchy.
- The DNS server searches for the record matching the requested domain name and returns the corresponding IP address to the browser.
- The browser uses that IP address to establish a connection to the target server.

Types of DNS Servers as illustrated Figure 6.1

- DNS Resolver** — the first point of contact; queries other servers on behalf of the client.
- Root Name Server** — the top of the DNS hierarchy; directs queries to the correct TLD server.
- TLD Name Server** — handles top-level domains such as .com, .org, or .id.
- Authoritative Name Server** — holds the actual DNS records for a domain and provides the final answer.

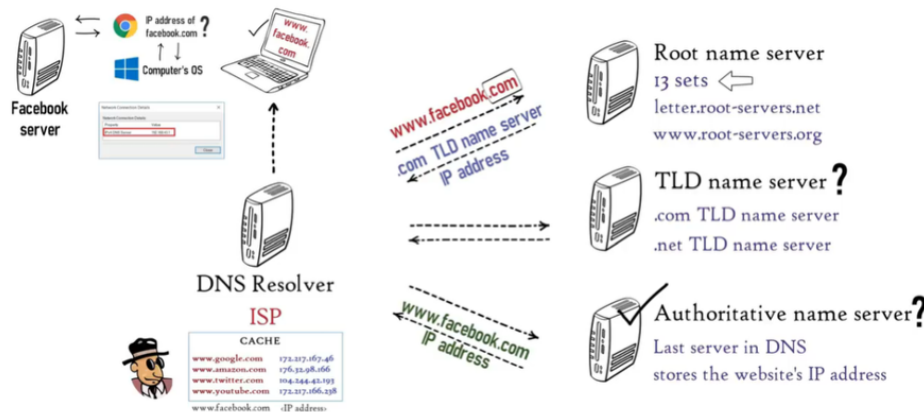


Figure 6.1: Types of DNS Servers

6.3 DNS Server Installation

! Before you begin

These steps assume Ubuntu Server is already installed, running, and reachable from the Windows host machine. If not, complete the Ubuntu Server installation practicum first.

6.4 1. Installation

Step 1 — Find the Ubuntu Server IP address and test connectivity

On Ubuntu Server, check the IP address:

```
ip a
```

①

① Note the IP address on the `enp0s8` adapter (e.g. `192.168.56.103`).

Then from Windows **Command Prompt**, ping that address and confirm you receive replies:

```
ping 192.168.56.103
```

Step 2 — Confirm the domain name does not yet resolve

Choose a domain name to use throughout this practicum (e.g. `yourdomain.com`). From Windows **Command Prompt**, ping it:

```
ping yourdomain.com
```

The expected output is:

```
Ping request could not find host yourdomain.com.  
Please check the name and try again.
```

This confirms no DNS record exists for the name yet.

Step 3 — Install BIND9

Back on Ubuntu Server, log in as root and update the package list:

```
apt update
```

Install the BIND9 DNS server:

```
apt-get install bind9
```

①

① BIND9 (Berkeley Internet Name Domain) is the most widely used DNS server software on Linux.

Verify the installation:

```
named -v  
systemctl status named
```

①

① Displays the BIND9 version; confirms the binary is installed correctly.

Allow BIND9 through the firewall:

```
ufw allow bind9
```

6.5 2. DNS Configuration

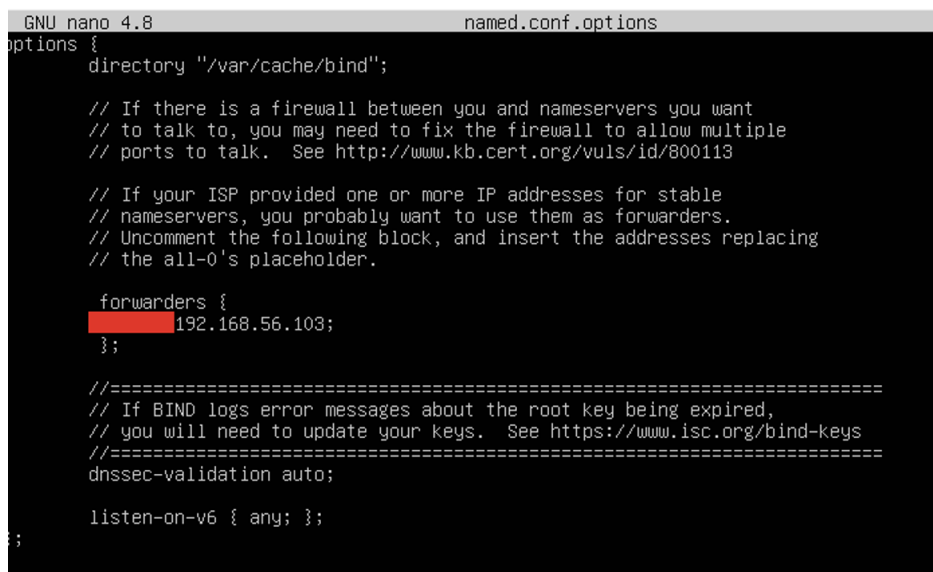
Step 4 — Configure forwarders in `named.conf.options`

```
cd /etc/bind/  
nano named.conf.options
```

Find the **forwarders** block, remove the double slashes (//) to uncomment it, and enter your DNS forwarder address:

```
forwarders {  
    8.8.8.8;  
};
```

Save and exit, as shown in Figure 6.2.



```
GNU nano 4.8 named.conf.options  
options {  
    directory "/var/cache/bind";  
  
    // If there is a firewall between you and nameservers you want  
    // to talk to, you may need to fix the firewall to allow multiple  
    // ports to talk.  See http://www.kb.cert.org/vuls/id/800113  
  
    // If your ISP provided one or more IP addresses for stable  
    // nameservers, you probably want to use them as forwarders.  
    // Uncomment the following block, and insert the addresses replacing  
    // the all-0's placeholder.  
  
    forwarders {  
        192.168.56.103;  
    };  
  
    //=====   
    // If BIND logs error messages about the root key being expired,  
    // you will need to update your keys.  See https://www.isc.org/bind-keys  
    //=====   
    dnssec-validation auto;  
  
    listen-on-v6 { any; };  
};
```

Figure 6.2: Forwarders configuration in named.conf.options

Step 5 — Declare zones in named.conf.local

```
nano named.conf.local
```

Append the following two zone declarations at the end of the file:

```
zone "yourdomain.com" {  
    type master;  
    file "/etc/bind/db.yourdomain1";  
};  
  
zone "56.168.192.in-addr.arpa" {  
    type master;  
    file "/etc/bind/db.yourdomain2";  
};
```

Save and exit.

Step 6 — Copy the zone file templates

```
cp db.local db.yourdomain1  
cp db.127 db.yourdomain2
```

①
②

- ① db.local is the template for a forward lookup zone.
- ② db.127 is the template for a reverse lookup zone.

Verify both files were created:

```
ls
```

Step 7 — Edit the forward zone file (db.yourdomain1)

```
nano db.yourdomain1
```

Replace the relevant portion of the file with the following:

```
@    IN    SOA    yourdomain.com.  root.yourdomain.com. (
                        2          ; Serial
                        604800     ; Refresh
                        86400      ; Retry
                        2419200    ; Expire
                        604800 )   ; Negative Cache TTL
;
@    IN    NS     yourdomain.com.
@    IN    A      192.168.56.103
www  IN    A      192.168.56.103
```

Save and exit.

Step 8 — Edit the reverse zone file (db.yourdomain2)

```
nano db.yourdomain2
```

Replace the relevant portion of the file with the following:

```
@    IN    SOA    yourdomain.com.  root.yourdomain.com. (
                        1          ; Serial
                        604800     ; Refresh
                        86400      ; Retry
                        2419200    ; Expire
                        604800 )   ; Negative Cache TTL
;
@    IN    NS     yourdomain.com.
103  IN    PTR    yourdomain.com.
```

Save and exit.

6.6 3. Network Setup

Step 9 — Validate the configuration

Check the BIND9 configuration syntax:

```
named-checkconf
```

①

① No output means no errors were found.

Validate each zone file:

```
named-checkzone yourdomain.com /etc/bind/db.yourdomain1
named-checkzone 56.168.192.in-addr.arpa /etc/bind/db.yourdomain2
```

If no errors appear, the configuration is correct.

Step 10 — Add the DNS server address to Netplan

```
nano /etc/netplan/01-netconf.yaml
```

Update the `enp0s8` section to include nameservers:

```
network:
  version: 2
  renderer: networkd
  ethernets:
    enp0s8:
      dhcp4: yes
      nameservers:
        addresses:
          - 8.8.8.8
          - 192.168.56.103
```

①

① This tells the Ubuntu Server to use itself as the primary DNS resolver.

Apply the new network configuration:

```
netplan apply
```

Step 11 — Restart BIND9

```
service bind9 restart
```

Step 12 — Configure the Windows network adapter DNS

On Windows, open the network adapter settings for the **VirtualBox Host-Only Ethernet Adapter** that connects to your VM. Set the **Preferred DNS server** to the Ubuntu Server's IP address (e.g. 192.168.56.103), as shown in Figure 6.3.

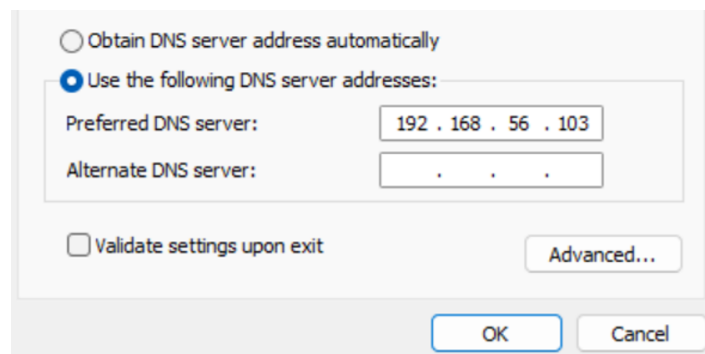


Figure 6.3: DNS adapter settings on Windows

6.7 Exercise

1. Run the following commands from the **Ubuntu Server** terminal and observe the results:

```
nslookup yourdomain.com
nslookup 192.168.56.103
dig yourdomain.com
ping yourdomain.com
```

①

②

③

2. From the Windows machine, open a web browser and navigate to:

`http://yourdomain.com`

If the Apache web server from the previous practicum is still running, the web page should load

successfully — confirming that DNS resolution is working end-to-end.

3. Based on your observations in Exercise 1, explain the function of each command:

Q1: What information does `nslookup` return, and in what situations would you use it?

Q2: What additional information does `dig` provide compared to `nslookup`?

CHAPTER VII PROXY SERVER

7.1 Practical Objectives

- a. Students are able to install a Proxy Server.
- b. Students are able to configure Squid Proxy Server, set the proxy on the client side, and block specific sites or URLs through the Proxy Server.

7.2 Theoretical Background

A proxy server acts as an intermediary between a user and the internet [14]. When a user wants to access a website or online service, the request is first sent to the proxy server, which forwards it to the destination. The response then travels back through the proxy before reaching the user.

This intermediary role provides several practical benefits:

Faster internet access — By caching frequently requested content locally, a proxy server reduces the time needed to retrieve it from the internet on subsequent requests.

Improved security — A proxy can filter traffic to block malicious content and network attacks. Some proxies also provide encryption to protect user privacy.

Bandwidth savings — Caching popular content reduces repeated transfers over the network, lowering bandwidth consumption and internet costs.

Access control — A proxy server can restrict or block access to specific websites. Conversely, it can also be used to reach sites that are otherwise inaccessible from the local network.

About Squid

Squid is a widely used open-source proxy and caching server for Linux. It supports HTTP, HTTPS, and FTP, and is configurable through a single file: `/etc/squid/squid.conf` [15].

7.3 Squid Workflow

7.4 1. Installation

Step 1 — Update the package list

```
apt-get update
```

Step 2 — Install Squid

```
apt-get install squid
```

①

① Squid is available in the Ubuntu package repository; no additional source is needed.

Step 3 — Verify the installation

```
systemctl status squid
```

①

① The output should show **active (running)** to confirm a successful installation.

Step 4 — Allow Squid through the firewall

```
ufw allow squid
```

①

① Opens port **3128** (Squid's default port) in the firewall so clients can connect.

7.5 2. Configuration

Step 1 — Navigate to the Squid directory

```
cd /etc/squid
ls
```

Step 2 — Back up the default configuration

```
cp squid.conf squid.conf.backup
```

①

① Always back up before editing so you can restore the original if needed.

Step 3 — Open the configuration file and note the port

```
nano squid.conf --linenumbers
```

①

① The `--linenumbers` flag displays line numbers, making it easier to navigate the file. Use `Ctrl+W` to search. Find the `http_port` line (around line 2179 if unchanged) and note the port number — it is typically **3128**.

Step 4 — Allow all traffic

Still in `squid.conf`, search for the comment `your own rule`. Find the line:

```
http_access deny all
```

Change it to:

```
http_access allow all
```

Save and exit with `Ctrl+X`, then `Y`, then `Enter`.

Step 5 — Restart Squid

```
service squid restart
```

7.6 Exercise

7.7 1. Client Proxy Setup

Configure the Windows client machine to route its traffic through the Squid proxy.

Step 1 — Open proxy settings

On the client machine, open a browser (e.g. Google Chrome) and navigate to the system proxy settings.

Opening proxy settings on Windows

Windows 11: **Settings** → **Network & internet** → **Proxy** → under *Manual proxy setup*, click **Edit** next to *Use a proxy server*.

Step 2 — Enable the proxy

In the **Edit proxy server** window:

- Toggle **Use a proxy server** to **On**.
- Enter the **Address**: the Ubuntu Server's IP address (e.g. `192.168.56.103`).
- Enter the **Port**: the port number noted in Configuration Step 3 (typically `3128`).
- Click **Save**.

Step 3 — Test the connection

Open a browser on the client and navigate to a website, for example `youtube.com`. The page should load normally, confirming that traffic is being routed through the proxy.

7.8 2. Blocking Sites

Block specific domains by creating an access control list (ACL) file that Squid reads.

Step 1 — Open `squid.conf`

```
nano /etc/squid/squid.conf
```

Step 2 — Add the ACL rules

Search for the `your own rule` comment. Directly below the line:

```
include /etc/squid/conf.d/*
```

Add the following two lines:

```
acl bad_url dstdomain "/etc/squid/bad_url.acl"  
http_access deny bad_url
```

Save and exit.

Step 3 — Create the blocklist file

```
nano /etc/squid/bad_url.acl
```

①

- ① Each line in this file is a domain pattern that Squid will block. A leading dot (.) matches the domain and all its subdomains.

Add the domains to block, one per line:

```
.yahoo.com  
.youtube.com  
.facebook.com
```

Save and exit.

Step 4 — Restart Squid and test

```
service squid restart
```

From the client browser, try to access one of the blocked domains. The browser should display a connection error or Squid's access denied page, confirming the block is active.

CHAPTER VIII SARG

8.1 Practical Objectives

- a. Students are able to install SARG on Ubuntu Server.
- b. Students are able to configure SARG to generate web access reports from Squid proxy logs.
- c. Students are able to read and interpret SARG-generated reports via a web browser.

8.2 Theoretical Background

SARG (Squid Analysis Report Generator) is an open-source tool that reads Squid proxy log files and produces detailed, human-readable HTML reports about network activity [16]. Rather than parsing raw logs manually, SARG automates the analysis and presents the data in an organised, browsable format.

Table 8.1: SARG report categories

Report Category	What It Shows
Traffic Statistics	Volume of requests and data transferred per user or site
Top Sites	Most frequently visited domains
Access Time Patterns	When users are most active on the network
Bandwidth Usage	Data consumption per user, site, or time period
Error Reports	Blocked requests, denied access events, and HTTP errors

Squid records all proxy activity to a log file — by default `/var/log/squid/access.log` — in a fixed columnar format that includes timestamp, response time, client IP, HTTP status, bytes transferred, and the requested URL [17]. SARG reads this file and transforms it into the categorised reports above.

8.3 Workflow SARG

8.3.1 SARG Installation

Log in to Ubuntu Server as root and update the package list:

```
apt update
```

Install SARG:

```
apt install sarg
```

①

① SARG is available in the Ubuntu package repository; no additional PPA is required.

8.3.2 SARG Configuration

All SARG settings live in a single configuration file: `/etc/sarg/sarg.conf`. The steps below make three targeted edits to that file, then integrate SARG reports into Apache.

8.4 Preparation

Step 1 — Navigate to the SARG configuration directory

```
cd /etc/sarg/
```

Step 2 — Back up the default configuration

```
cp sarg.conf sarg.conf.bak
```

①

① Always back up before editing so you can restore the original if needed.

8.5 Editing sarg.conf

Step 3 — Open the configuration file

```
nano sarg.conf
```

Make the four edits in Table 8.2, then save and exit.

Table 8.2: Key edits in sarg.conf

Line	Action	Before	After
120	Comment out	output_dir /var/www/html/squid-reports	#output_dir /var/www/html/squid-reports
121	Uncomment	#output_dir /var/www/sarg	output_dir /var/www/sarg
132	Set to yes	resolve_ip no	resolve_ip yes
377	Set charset	#charset UTF-8	charset UTF-8

What these settings do

- **Lines 120–121** redirect report output to `/var/www/sarg`, which Apache can serve directly.
- **Line 132** (`resolve_ip yes`) attempts to resolve client IP addresses to hostnames in the report.
- **Line 377** sets the HTML character encoding to UTF-8, ensuring correct rendering of non-ASCII characters.

8.6 Apache Integration

Step 4 — Create the report output directory

```
mkdir /var/www/sarg
```

Step 5 — Create the Apache configuration file for SARG

```
nano /etc/apache2/conf-available/sarg.conf
```

Enter the following content:

```
Alias /squid-reports /var/www/sarg

<Directory /var/www/sarg>
    Options Indexes FollowSymLinks
    AllowOverride None
    Require all granted
</Directory>
```

Save and exit.

Step 6 — Enable the configuration and reload Apache

```
a2enconf sarg
systemctl reload apache2
```

①

- ① `a2enconf` creates a symlink from `conf-enabled/sarg.conf` to `conf-available/sarg.conf`, making the configuration active.

8.7 Exercise

Complete the following steps in order. Steps 1–2 generate proxy log entries on the client; steps 3–5 set permissions and run SARG; step 6–7 view and interpret the results.

Step 1 — Connect the client to the proxy

On the client machine (Windows), configure the browser to use the Ubuntu Server IP address and port **3128** as the HTTP proxy (this is the Squid proxy configured in the previous practicum).

Step 2 — Generate log entries

Browse to several different websites from the client. These requests will be recorded in Squid's access log and will appear in the SARG report.

Step 3 — Set permissions on the report directory

Back on Ubuntu Server, grant write access to the SARG output directory:

```
chmod 775 /var/www/sarg
```

①

- ① SARG writes the HTML report files here; it needs write permission to do so.

Step 4 — Set permissions on the Squid log

```
chmod 775 /var/log/squid/access.log
```

①

- ① SARG reads this file; it needs read permission on the log.

Step 5 — Run SARG to generate the report

```
/usr/bin/sarg
```

①

- ① SARG reads `/var/log/squid/access.log`, processes all entries, and writes the HTML report to `/var/www/sarg`.

Step 6 — View the report in a browser

From the Windows machine, open a browser and navigate to:

`http://192.168.56.101/squid-reports/`

A directory listing of report folders (organised by date) should appear, as shown in Figure 8.1.

Step 7 — Explore the report sections

Open the most recent report folder and browse through the report. Identify the following sections and note what each one shows:

- **TOP** — the most-visited websites during the report period
- **USERS** — per-user activity breakdown
- **SITES** — all sites accessed, with request counts and bytes transferred

Q: Write a brief description of what you observed in each section based on the browsing activity you generated in Step 2.

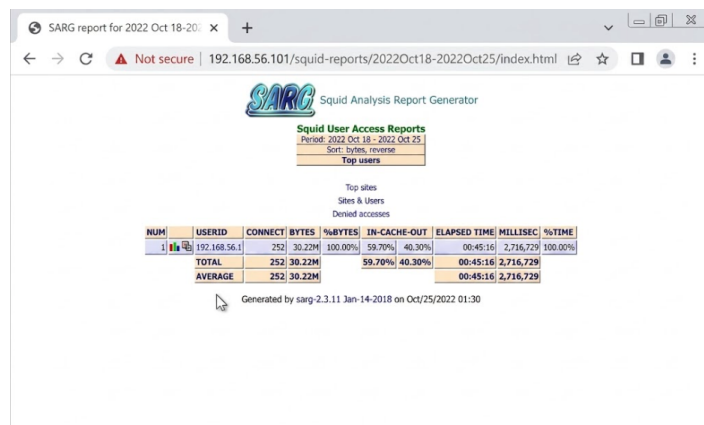


Figure 8.1: Directory listing of SARG reports

CHAPTER IX MAIL SERVER

9.1 Practical Objectives

- a. Students are able to install a Mail Server using Postfix and Dovecot.
- b. Students are able to manage mail user accounts and demonstrate email delivery through the Mail Server.

9.2 Theoretical Background

A mail server is a system responsible for receiving, storing, and sending email over a computer network. It acts as the central hub for email exchange between senders and recipients.

How email delivery works:

1. The user composes and sends a message using a mail client.
2. The message travels from the user's host to the local mail server.
3. The local mail server relays the message to the destination mail server over the internet.
4. The destination mail server delivers the message to the recipient's mailbox.

9.2.1 Email Agents

Three types of agents are involved in the email delivery process:

Full		
Agent	Name	Role
MUA	Mail User Agent	The mail client used by the user to compose and send email (e.g. Thunderbird, Outlook)
MTA	Mail Transfer Agent	The mail server that uses SMTP to relay messages between servers (e.g. Postfix)
MDA	Mail Delivery Agent	Retrieves incoming messages from the MTA and delivers them to the user's local mailbox

Figure 9.1 is a diagram illustrating the three types of agents involved in the email delivery process.

9.2.2 Protocols

SMTP (Simple Mail Transfer Protocol) is the standard protocol for sending email between mail servers [18]. Key characteristics:

- Uses **port 25** by default; port **587** is commonly used for authenticated client submission.
- Requires username and password authentication to send mail through a relay server.

POP3 and IMAP are the two protocols used to retrieve email from a mail server:

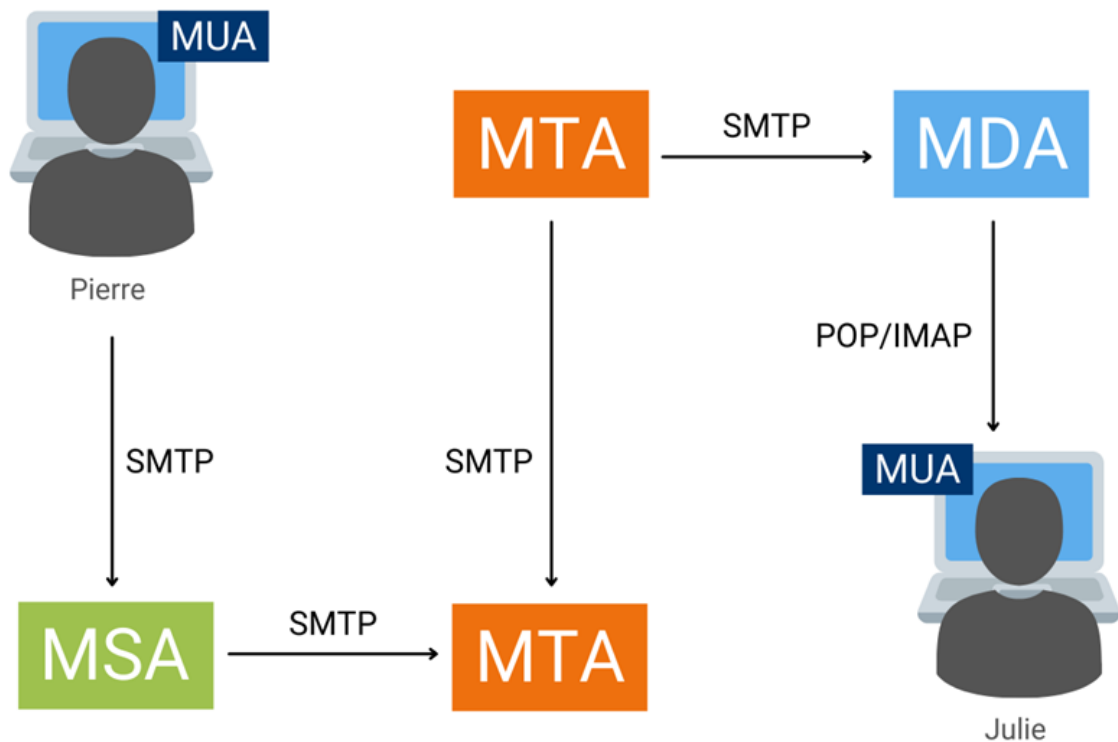


Figure 9.1: Email Delivery Process

Feature	POP3	IMAP
Storage	Email downloaded to device; removed from server	Email stays on server
Multi-device access	No — email lives on one device	Yes — synced across all devices
Offline access	Yes, after downloading	Limited to cached content
Folder synchronization	No	Yes
Default port	110	143

POP3 suits users who want to download email once and read it offline. IMAP suits users who need to access their mailbox from multiple devices with full synchronization.

9.3 Mail Server Installation

9.4 1. Preparation

Step 1 — Log in as root and update

```
apt-get update
```

Step 2 — Check the current hostname

```
hostnamectl status
```

Note the existing hostname — you will need to replace it in `/etc/hosts` in Step 4.

Step 3 — Set the mail server hostname

```
hostnamectl set-hostname mail.example.com
```

Step 4 — Update `/etc/hosts`

```
nano /etc/hosts
```

On line 2, replace the old IP address and hostname with your Ubuntu Server's IP address and the new hostname set in Step 3:

```
192.168.56.103    mail.example.com
```

Save and exit.

Step 5 — Reboot

```
reboot
```

9.5 2. Postfix Setup

Step 1 — Install Postfix, Dovecot, and supporting packages

After rebooting, log back in as root, then run:

```
apt -y install postfix sasl2-bin dovecot-core dovecot-pop3d dovecot-imapd mailutils ①
```

- ① When the package configuration prompt appears, select **No Configuration** — Postfix will be configured manually in the next steps.

Step 2 — Copy the Postfix configuration template

```
cp /usr/share/postfix/main.cf.dist /etc/postfix/main.cf
```

Step 3 — Edit `main.cf`

```
nano /etc/postfix/main.cf --linenumbers ①
```

- ① The `--linenumbers` flag displays line numbers in the editor. Use `Ctrl+W` to search for a line's content.

Apply the edits in Table 9.3. Replace `192.168.56.103` with your actual Ubuntu Server IP address where applicable.

Table 9.3: Edits to `/etc/postfix/main.cf`

Line	Action	Value
82	Uncomment	<code>mail_owner = postfix</code>
98	Uncomment & edit	<code>myhostname = mail.example.com</code>

Line	Action	Value
106	Uncomment & edit	mydomain = example.com
127	Uncomment	myorigin = \$mydomain
141	Uncomment	inet_interfaces = all
189	Uncomment	mydestination = \$myhostname, localhost.\$mydomain, localhost, \$mydomain
232	Uncomment	local_recipient_maps = unix:passwd.byname \$alias_maps
277	Uncomment	mynetworks_style = subnet
294	Add server IP	mynetworks = 127.0.0.0/8, 192.168.56.103/24
416	Uncomment	alias_maps = hash:/etc/aliases
427	Uncomment	alias_database = hash:/etc/aliases
449	Uncomment	home_mailbox = Maildir/
585	Comment out	#smtpd_banner = \$myhostname ESMTP \$mail_name (Ubuntu)
586	Add below 585	smtpd_banner = \$myhostname ESMTP
659	Add	sendmail_path = /usr/sbin/postfix
664	Add	newaliases_path = /usr/bin/newaliases
669	Add	mailq_path = /usr/bin/mailq
675	Add	setgid_group = postdrop
679	Comment out	#html_directory =
683	Comment out	#manpage_directory =
688	Comment out	#sample_directory =
692	Comment out	#readme_directory =
693	Set to ipv4	inet_protocols = ipv4

At the **end of the file**, append the following block:

```
message_size_limit = 10485760
mailbox_size_limit = 1073741824

# SMTP-Auth settings
smtpd_sasl_type = dovecot
smtpd_sasl_path = private/auth
smtpd_sasl_auth_enable = yes
smtpd_sasl_security_options = noanonymous
smtpd_sasl_local_domain = $myhostname
smtpd_recipient_restrictions = permit_mynetworks, permit_auth_destination, permit_sasl_authentication
```

Save and exit.

Step 4 — Rebuild the alias database

```
newaliases
```

9.6 3. Dovecot Setup

Step 5 — Edit dovecot.conf

```
nano /etc/dovecot/dovecot.conf --linenumbers
```

At line 30, uncomment:

```
listen = *, ::
```

Save and exit.

Step 6 — Edit 10-auth.conf

```
nano /etc/dovecot/conf.d/10-auth.conf --linenumbers
```

Line	Change
10	Uncomment and set to no: disable_plaintext_auth = no
100	Set to: auth_mechanisms = plain login

Save and exit.

Step 7 — Edit 10-mail.conf

```
nano /etc/dovecot/conf.d/10-mail.conf --linenumbers
```

At line 30, change to:

```
mail_location = maildir:~/Maildir
```

Save and exit.

Step 8 — Edit 10-master.conf

```
nano /etc/dovecot/conf.d/10-master.conf --linenumbers
```

At lines 109–112, uncomment the `unix_listener` block and add the user and group:

```
# Postfix smtp-auth
unix_listener /var/spool/postfix/private/auth {
  mode = 0666
  user = postfix
  group = postfix
}
```

Save and exit.

Step 9 — Restart services and verify

```
systemctl restart dovecot postfix
systemctl status dovecot postfix
```

Confirm that Dovecot shows **active (running)** and Postfix shows **active (exited)**.

9.7 Exercise

9.8 Add Mail User

Step 1 — Set the mail environment variable

```
echo 'export MAIL=$HOME/Maildir/' >> /etc/profile.d/mail.sh
```

①

① This ensures the `mail` command knows where to look for each user's mailbox.

Step 2 — Note the existing username

Run the following to see the current user accounts on the system:

```
cat /etc/passwd | grep home
```

①

- ① Identify your primary Ubuntu user account (e.g. `alam`). This will be the email recipient in the next exercise.

Step 3 — Add a new user

```
adduser ubuntu
```

①

- ① Follow the prompts to set a password. Leave Full Name, Room Number, and other fields blank by pressing Enter.

After the user is created, reboot the server:

```
reboot
```

9.9 Send Test Email

Step 4 — Send an email from the new user

After rebooting, switch to the new user account:

```
su ubuntu
```

Send an email to the original user (e.g. `alam`):

```
mail alam@localhost
```

Follow the prompts:

- **Cc:** press Enter to skip.
- **Subject:** type a subject, e.g. `Test email 1`, then press Enter.
- **Body:** type the email body, e.g.:
`This is the first test email.`
`Thank you.`
- Press Ctrl+D to finish and send.

Step 5 — Switch back to the original user

```
exit
```

①

- ① If you are still in the root shell, type `exit` a second time to return to the original user account. A notification should appear indicating that new mail has arrived.

Step 6 — Read the email

Open the mailbox:

```
mail
```

Type the message number (e.g. `1`) and press Enter to read it. Type `q` and press Enter to quit.

CHAPTER X WEBMIN

10.1 Practical Objectives

- a. Students are able to install Webmin on Ubuntu Server.
- b. Students are able to use Webmin features to perform system administration and network management tasks.

10.2 Theoretical Background

Webmin is an open-source, web-based administration interface for Linux and Unix systems, developed by Jamie Cameron [19]. Rather than requiring administrators to memorise shell commands, Webmin provides a graphical interface accessible from any web browser — making system management more approachable for users who are not yet fluent with the Linux terminal.

A note on system administration

Webmin makes powerful operations easy to trigger. Always make sure you understand what an action does before applying it — mistakes in system configuration can be difficult to reverse and may affect server stability.

Webmin’s capabilities are organised into modules. The core modules cover the following areas:

Table 10.1: Webmin core module areas

Module area	What you can do
User & Group Management	Add, edit, and delete user accounts and groups; manage permissions and access rights
Service Management	Start, stop, restart, and configure services such as Apache, MySQL, and SSH
Network Configuration	Set IP addresses, manage network interfaces, and adjust network settings
Package Management	Install, remove, and update software packages through a web interface
Disk & File Management	Manage disk partitions, storage allocation, and file system operations
Security Configuration	Set file permissions, configure the firewall, and apply other security measures

Webmin supports a wide range of Linux distributions and can be extended further through community-developed modules that add support for additional applications and configurations.

10.3 Webmin Installation

Log in to Ubuntu Server as root, then follow the steps below.

Step 1 — Download the repository setup script

```
curl -o webmin-setup-repo.sh https://raw.githubusercontent.com/webmin/webmin/master/webmin-setup-r
```

Step 2 — Run the script to add the repository and GPG key

```
sudo sh webmin-setup-repo.sh
```

①

- ① This adds the official Webmin package repository and its signing key to the system, so the package can be installed and updated through `apt`.

Step 3 — Update the package list

```
apt-get update
```

Step 4 — Install Webmin

```
apt install webmin --install-recommends -y
```

①

- ① The `--install-recommends` flag installs optional but recommended packages that enable additional Webmin features.

Step 5 — Open port 10000 in the firewall

```
ufw allow 10000/tcp
```

①

- ① Webmin listens on port **10000** over HTTPS. This rule allows incoming connections to that port.

10.4 Exercise

10.5 Explore Webmin

Step 1 — Access the Webmin dashboard

From the Windows host machine, open a browser and navigate to:

`https://192.168.56.103:10000`

Replace `192.168.56.103` with your Ubuntu Server's IP address.

Security warning

If the browser displays a **Potential Security Risk Ahead** warning, click **Advanced** → **Accept the Risk and Continue**. This warning appears because Webmin uses a self-signed certificate, which browsers do not trust by default.

Step 2 — Log in

Enter your Ubuntu Server's root username and password, then click **Sign In** as shown Figure 10.1.

Step 3 — Explore the dashboard

Once logged in, observe the information displayed on the Webmin dashboard. Answer the following question:

Q: What system information is shown on the dashboard? List the categories and describe what each one tells you about the server.

Step 4 — Explore one feature in depth

Navigate through the Webmin menu and choose **one feature** that you consider important. Answer:

Q: Which feature did you select, under which menu does it appear, and what does it allow an administrator to do?

Step 5 — Explore four more features

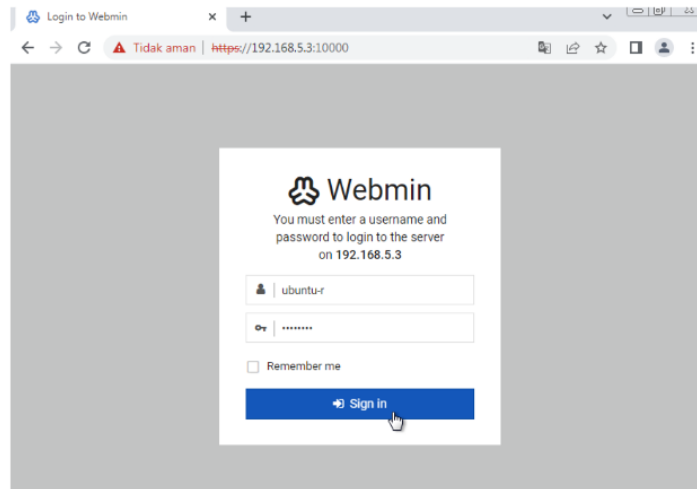


Figure 10.1: Webmin login screen

Browse through the remaining menus and identify **four additional features** that you consider useful or interesting. For each one, state the feature name, its menu location, and its purpose.

Step 6 — Test the reboot function

Navigate to **System → Bootup and Shutdown**. Use the interface to reboot the Ubuntu Server. Confirm that the server comes back online and that Webmin is accessible again after the reboot.

10.6 Add Account

Webmin has its own user system, separate from Linux system accounts. A Webmin user can be granted access to specific modules without having a full Linux login.

Step 1 — Open the Webmin Users module

In the Webmin menu, navigate to **Webmin → Webmin Users**.

Step 2 — Create a new privileged user

Click **Create a new privileged user**. Fill in the required fields:

- **Username** — a login name for the new Webmin user
- **Password** — set and confirm a password
- **Modules** — select which Webmin modules this user can access

Click **Save** to create the account.

Step 3 — Verify the new account

Log out of Webmin, then log back in using the new account's credentials. Confirm that the user can only access the modules that were assigned.

Part III

Projects

PROJECT 1

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